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Inventory Demand Forecasting via Local State-Space Kalman Filtering and Global Gradient Boosted Decision Trees

Comprehensive Phase 1 Evaluation on the Walmart M5 Dataset

Applied Machine Learning (CSAI) – Project 5
Team 5

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Codebase: `https://github.com/Jgauri24/aml-project`

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Abstract

Accurate multi-horizon inventory demand forecasting is vital for retail supply-chain efficiency, working-capital optimisation, and stockout reduction. Retail sales series are difficult to model: they are highly intermittent (69.33% of daily records are zero), show a pronounced weekly cycle, and react to promotional shocks. This report compares two contrasting forecasting paradigms — local state-space tracking (five Kalman-filter variants) and a global, direct multi-horizon LightGBM model — against two seasonal heuristic baselines. Experiments use a deterministic, stratified cohort of 240 Walmart M5 series and an expanding four-fold rolling-origin backtest with a 28-day horizon (26,880 out-of-sample predictions); the final competition window is kept sealed. Global LightGBM achieves the best point accuracy (MAE 0.9227, WAPE 0.7640). The Kalman local-level model with weekly seasonality achieves the lowest RMSE (2.0535) and, like the other Kalman variants, produces closed-form 80% prediction intervals (81.08% empirical coverage for the single-SKU model and 85–86% for the other variants). These results motivate a dual-model operating policy: LightGBM for procurement sizing and Kalman predictive variance for safety stock.

Keywords: demand forecasting, state-space models, Kalman filter, maximum likelihood estimation, gradient boosting, LightGBM, rolling-origin backtesting, uncertainty quantification, M5 competition.

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1 Introduction

Modern supply-chain operations need forecasting systems that support two distinct decisions:

1. **Point demand sizing** — accurate conditional-mean or median estimates used to size procurement batches, warehouse logistics, and replenishment orders.
2. **Safety-stock sizing** — calibrated prediction intervals and variance bounds that cushion against demand volatility without inflating holding costs.

Retail daily sales, as in the Walmart M5 dataset [5], violate many textbook assumptions. They are intermittent (many zero-demand days), show asymmetric promotional spikes (for example, Supplemental Nutrition Assistance Program (SNAP) disbursement days), have stepped price changes, and follow a 7-day calendar cycle.

Practitioners typically choose between two families of methods:

- **Local time-series models** such as ARIMA, Croston's method and state-space Kalman filters [1, 3]. Parameters are fitted independently per series; they adapt quickly to level shifts and handle missing observations natively, but cannot share structure across sparse series.
- **Global tabular machine-learning models** such as gradient boosted decision trees (LightGBM [4]) and deep networks [2]. They pool histories from many items so that slow-moving products can borrow statistical strength from fast-moving ones.

1.1 Objectives and Contributions

This Phase 1 study benchmarks local Kalman filters against a global LightGBM model on a fixed cohort of 240 M5 series. Our contributions are:

1. a reproducible, stratified 240-series cohort and a cutoff-safe exploratory analysis (no look-ahead);
2. five per-series Kalman state-space variants with maximum-likelihood hyper-parameter estimation and closed-form predictive quantiles;
3. a leakage-safe 26-feature pipeline and a direct multi-horizon LightGBM forecaster trained with an L1 loss;
4. a four-fold rolling-origin backtest (26,880 predictions) evaluated with point, error-ratio and probabilistic metrics; and
5. an operational recommendation that combines the strengths of both paradigms.

2 Dataset and Cutoff-Safe Exploratory Analysis

2.1 Walmart M5 Dataset and Cohort Selection

The data source is the Walmart M5 hierarchical sales dataset: daily unit sales for items in stores across California (CA), Texas (TX) and Wisconsin (WI). To keep experiments reproducible and tractable, a deterministic, stratified cohort of $N = 240$ series is built with a fixed seed (42) by `src/data/select_skus.py`:

- **Volume stratification:** exactly 80 low-, 80 medium- and 80 high-volume series, split by trailing-365-day sales up to the cutoff day `d1801`.
- **Hierarchical diversity:** the cohort spans all 3 states, all 10 stores, all 3 categories (FOODS, HOUSEHOLD, HOBBIES) and all 7 departments.
- **No artificial selection bias:** no minimum-positive-sales rule is applied, so genuinely intermittent series are retained.

2.2 Key Findings of the Exploratory Analysis

To avoid look-ahead bias, all exploratory analysis uses only days `d1`–`d1801` (2011-01-29 to 2016-01-03; 432,240 series-days).

Table 1. Summary of cutoff-safe exploratory findings.

Aspect	Finding
Zero inflation	69.33% of all daily records are exact zeros; 89.06% in the low-volume group (median CV 3.27; longest zero streak 1,640 days).
Demand pattern (Syntetos–Boylan)	175 intermittent, 54 lumpy, 8 smooth, 3 erratic series.
Weekly seasonality	ACF peaks at lags 7, 14, 21, 28; median $r_7 = 0.1639$ (up to 0.3164 for high-volume SKUs); weekend-to-weekday peak-to-trough ratio 1.369×.
SNAP effect	State-specific lifts in the first 10 days of each month: CA +4.8%, TX +6.0%, WI +8.0%.
Price dynamics	Stepped, infrequent reductions; median price–demand correlation −0.0471; temporary rollbacks cause sharp spikes.

Because the share of zero actuals is so high, percentage metrics such as MAPE are undefined or diverge when $y_t = 0$. We therefore evaluate with the Weighted Absolute Percentage Error (WAPE) and the Mean Absolute Error (MAE) instead.

3 State-Space Modelling and Kalman Filtering

State-space models treat observed daily sales y_t as a noisy reflection of an unobserved demand state $\boldsymbol{\alpha}_t$.

3.1 Linear Gaussian State-Space System

$$y_t = Z_t \boldsymbol{\alpha}_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, R), \tag{1}$$

$$\boldsymbol{\alpha}_t = T_t \boldsymbol{\alpha}_{t-1} + \boldsymbol{\eta}_t, \qquad \boldsymbol{\eta}_t \sim \mathcal{N}(0, Q), \tag{2}$$

where Z_t is the observation vector, T_t the transition matrix, $R > 0$ the observation variance and $Q \succeq 0$ the process-noise covariance.

3.2 Evaluated Kalman Variants

Five per-series models were developed (Milestones 6 and 7):

Table 2. The five Kalman state-space variants.

Model	State	Description
`kalman_single_sku`	$[\mu_t]$	Milestone-6 reference local-level model; $T=[1]$, $Z=[1]$.
`kalman_local_level`	$[\mu_t]$	Random-walk level: $y_t=\mu_t+\varepsilon_t,\ \mu_t=\mu_{t-1}+\eta_t$.
`kalman_local_level_seasonal`	7-dim	Level plus a 6-dimensional weekly seasonal vector $\boldsymbol{\gamma}_t$ with $\sum_{j=0}^{6}\gamma_{t-j}=0$.
`kalman_local_trend`	$[\mu_t,\beta_t]$	Level plus damped slope ($\rho=0.98$), $T=\left(\begin{smallmatrix}1&1\\0&\rho\end{smallmatrix}\right)$, $Z=[1,0]$.
`kalman_local_trend_seasonal`	8-dim	Damped trend combined with weekly seasonality.

For the seasonal variants the state is $\boldsymbol{\alpha}_t=[\mu_t,\gamma_{t,1},\dots,\gamma_{t,6}]^\top$ with $Z=[1,1,0,0,0,0,0]$. The 6×6 seasonal transition block has ones on the superdiagonal and a final row of -1's, which enforces the zero-sum constraint above.

3.3 Recursive Filtering and Missing Data

Let $a_{t|t-1}=\mathbb{E}[\boldsymbol{\alpha}_t\mid\mathcal{F}_{t-1}]$ and $P_{t|t-1}=\mathrm{Var}(\boldsymbol{\alpha}_t\mid\mathcal{F}_{t-1})$. With a diffuse prior $P_0=10^4 I$, the filter proceeds as follows.

Time update (predict):

$$a_{t|t-1}=T\,a_{t-1|t-1},\qquad P_{t|t-1}=T\,P_{t-1|t-1}\,T^\top+Q. \tag{3}$$

Measurement update (correct), when y_t is observed:

$$v_t=y_t-Z a_{t|t-1} \quad \text{(innovation)} \tag{4}$$

$$S_t=Z P_{t|t-1}Z^\top+R \quad \text{(innovation variance)} \tag{5}$$

$$K_t=P_{t|t-1}Z^\top S_t^{-1} \quad \text{(Kalman gain)} \tag{6}$$

$$a_{t|t}=a_{t|t-1}+K_t v_t,\qquad P_{t|t}=(I-K_t Z)P_{t|t-1}. \tag{7}$$

Missing-data invariant. If y_t is unobserved (`NaN`), the update is skipped, $a_{t|t}=a_{t|t-1}$ and $P_{t|t}=P_{t|t-1}$: the filter coasts on its transition model rather than fabricating zeros.

3.4 Parameter Estimation by Maximum Likelihood

The hyper-parameters $\theta=(\ln Q,\ln R)$ are estimated per series by minimising the exact prediction-error negative log-likelihood

$$\ell(\theta)=\frac{1}{2}\sum_{t=1}^{T}\left[\ln(2\pi S_t)+\frac{v_t^2}{S_t}\right], \tag{8}$$

using L-BFGS-B with multiple restarts in log-variance space ($\theta\in[-10,8]$) to reduce the risk of poor local minima.

3.5 Multi-Step Predictive Distribution

For horizon $h \in \{1, \ldots, H\}$ from origin T (local-level case):

$$\mathbb{E}[y_{T+h} \mid \mathcal{F}_T] = \mu_{T|T}, \tag{9}$$

$$V_h = \mathrm{Var}(y_{T+h} \mid \mathcal{F}_T) = P_{T|T} + h\,Q + R, \tag{10}$$

so the predictive variance grows linearly with the horizon. Quantiles are clipped at zero because sales cannot be negative:

$$\hat{q}_\tau(y_{T+h}) = \max\!\Big(0,\ \hat{y}_{T+h} + z_\tau \sqrt{V_h}\Big), \qquad z_{0.10} = -1.282,\ z_{0.90} = +1.282. \tag{11}$$

4 Feature Engineering and Global LightGBM

4.1 Cross-Series Learning

Whereas each Kalman filter learns its own parameters, the global model pools all 240 series into one tabular training set. This lets low-velocity products borrow promotional, calendar and seasonal effects learned from dense items.

4.2 Leakage-Safe Feature Pipeline (Milestone 8)

`src/features/feature_engineering.py` builds 26 features anchored strictly at the forecast origin T (only information with $t \le T$):

- **Demand lags:** sales at $T-1$, $T-7$, $T-14$, $T-28$.
- **Rolling statistics:** 7-day and 28-day rolling means and standard deviations over $[T-W, T]$.
- **Calendar encodings:** day-of-week, day-of-month, month, weekend flag and circular sine/cosine transforms.
- **Exogenous indicators:** state SNAP flags (`snap_CA`, `snap_TX`, `snap_WI`), national event flags and sell price.
- **Direct-horizon invariant:** for horizon h, any lag feature with lag $< h$ is set to `NaN` and handled by LightGBM's native missing-value branching. A dedicated check function (`assert_no_future_leakage()`) enforces this.

4.3 LightGBM Forecaster (Milestone 9)

- **Loss:** L1 (`regression_l1`), so the trees target the conditional median, which is robust to zero inflation and holiday spikes.
- **Tuning:** rolling-validation tuning over the first three backtest folds selected a learning rate of 0.05, 500 estimators, 31 leaves per tree and a feature fraction of 0.8.

5 Experimental Protocol

5.1 Four-Fold Chronological Backtest

We use an expanding-window rolling-origin backtest with horizon $H = 28$ days (Figure 1). Each fold contains $240 \times 28 = 6{,}720$ forecasts, giving 26,880 out-of-sample predictions in total. The final competition period (`d1914`–`d1941`) is kept as a sealed holdout and is never used.

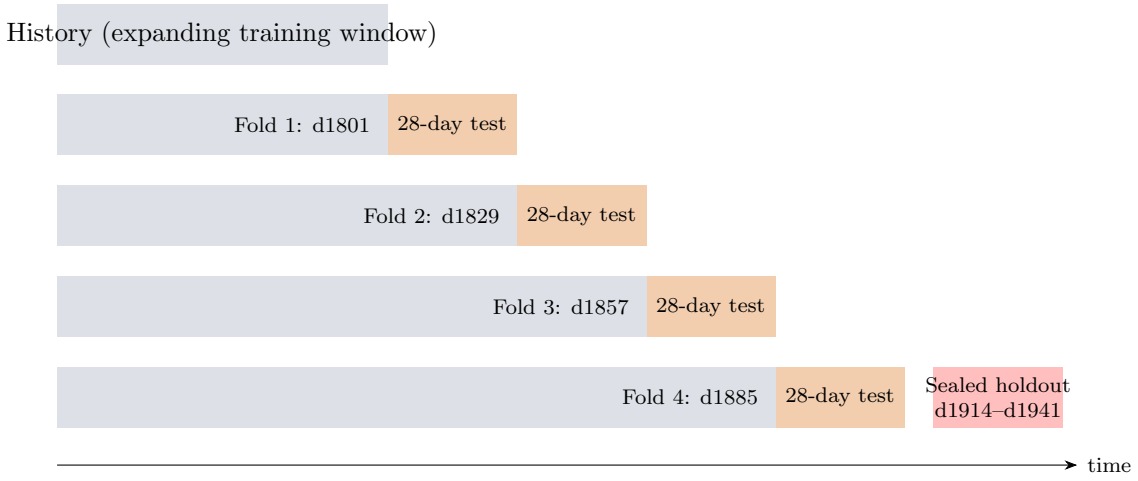


Figure 1. Expanding-window rolling-origin backtest. Each origin forecasts the next 28 days; the final competition window is never touched.

Table 3. Backtest folds.

Fold	Origin	Test window	Forecasts
1	`d1801`	2016-01-04 to 2016-01-31	6,720
2	`d1829`	2016-02-01 to 2016-02-28	6,720
3	`d1857`	2016-02-29 to 2016-03-27	6,720
4	`d1885`	2016-03-28 to 2016-04-24	6,720

5.2 Evaluation Metrics

With M evaluated points:

$$\mathrm{MAE} = \frac{1}{M}\sum_i |y_i - \hat{y}_i|, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{M}\sum_i (y_i - \hat{y}_i)^2}, \tag{12}$$

$$\mathrm{WAPE} = \frac{\sum_i |y_i - \hat{y}_i|}{\sum_i y_i}, \qquad \mathrm{PL}_\tau = \max\big(\tau(y - \hat{q}_\tau),\, (1-\tau)(\hat{q}_\tau - y)\big). \tag{13}$$

The empirical coverage PICP_{80} is the share of actuals falling in $[\hat{q}_{0.10}, \hat{q}_{0.90}]$, with a nominal value of 80%. Pinball loss and coverage are reported only for the Kalman models, which produce predictive quantiles.

6 Results

6.1 Pooled Point-Forecast Accuracy

Table 4 and Figure 2 summarise pooled accuracy over all 26,880 predictions.

Table 4. Point-forecast accuracy across all folds (26,880 series-days). Best value per column in green.

Model	Paradigm	MAE ↓	RMSE ↓	WAPE ↓
LightGBM (global)	Global tabular GBDT	**0.9227**	2.1081	**0.7640**
Kalman LL Seasonal	Local state-space	1.0007	**2.0535**	0.8286
Kalman LT Seasonal	Local state-space	1.0030	2.0576	0.8305
Kalman Local Level	Local state-space	1.0106	2.0613	0.8368
Kalman Single-SKU (M6)	Local state-space	1.0115	2.0776	0.8375
Kalman Local Trend	Local state-space	1.0225	2.0976	0.8466
Naive Weekly Seasonal	Heuristic baseline	1.1869	2.5487	0.9828
Naive Last Value	Heuristic baseline	1.2479	2.5151	1.0332

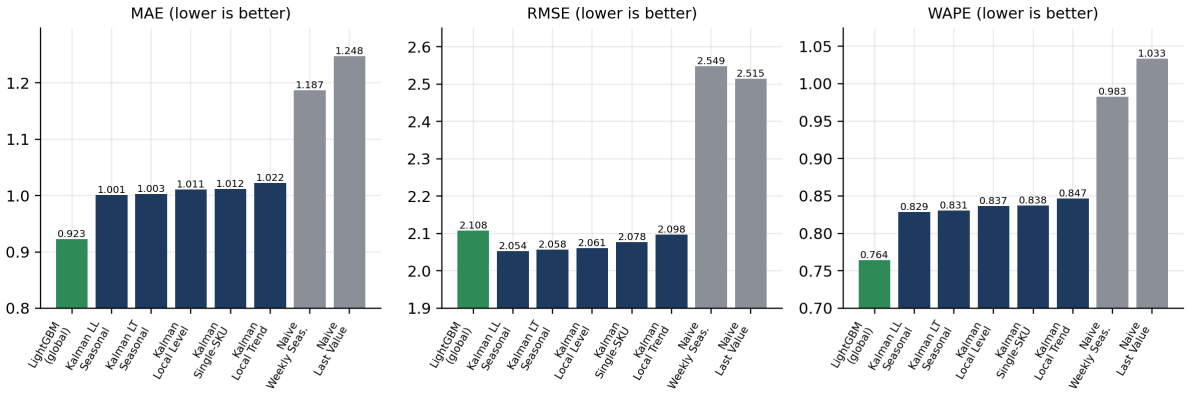


Figure 2. MAE, RMSE and WAPE for all eight models (LL = local level, LT = local trend). LightGBM (green) leads on MAE and WAPE; the seasonal local-level Kalman model leads on RMSE.

Main observations

- **Point accuracy:** LightGBM reduces MAE by 7.8% relative to the best Kalman model (1.0007) and by 22.3% relative to the best naive baseline, weekly seasonal (1.1869); against the last-value baseline the reduction is 26.1%.
- **Squared error:** the seasonal local-level Kalman model has the lowest RMSE (2.0535 vs. 2.1081 for LightGBM). RMSE penalises large errors heavily, so this suggests the Kalman forecasts make fewer large misses on high-demand days, although the pooled results alone do not prove the cause.
- **Trend states do not help:** the local-trend model has the worst Kalman WAPE (0.8466). Retail demand oscillates around a level rather than drifting steadily, so a trend component adds variance without benefit.

6.2 Comparison of the Kalman Variants

The five Kalman variants are close in pooled WAPE (0.8286–0.8466, Figure 3): adding weekly seasonality helps slightly (local level 0.8368 → 0.8286), whereas adding a damped trend hurts (0.8466). The differences are small compared with the gap to LightGBM or to the naive baselines.

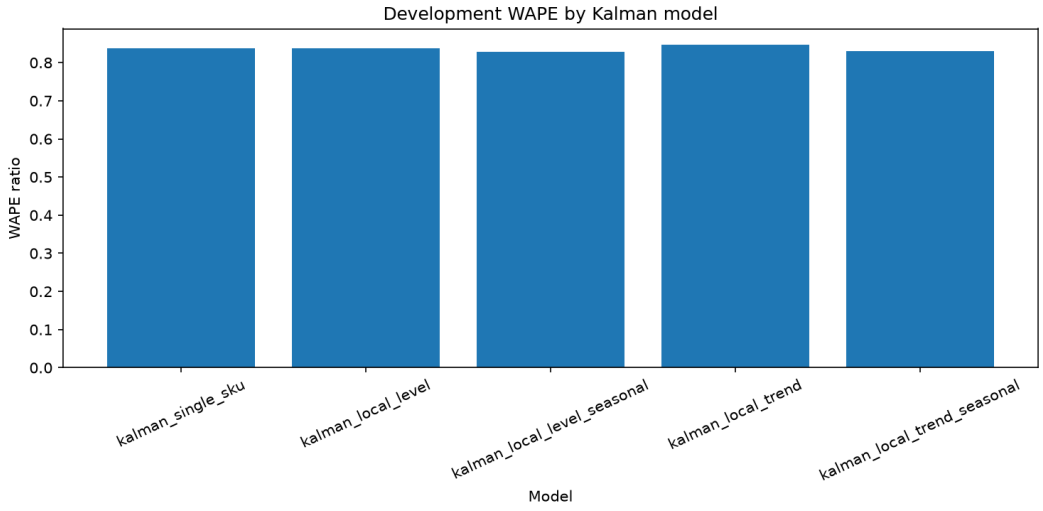


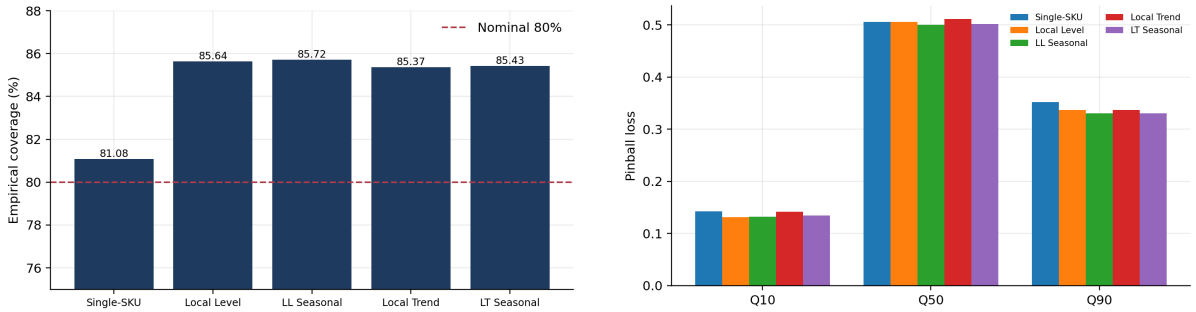
Figure 3. Development WAPE of the five Kalman variants. The bars are nearly equal in height; the seasonal local-level model is marginally lowest.

6.3 Probabilistic Forecasts: Pinball Loss and Coverage

Only the Kalman models produce predictive intervals. Table 5 and Figures 4a–4b show that every variant covers at least the nominal 80%. The single-SKU model is closest to nominal (81.08%), whereas the other four variants over-cover at roughly 85–86%, i.e. their intervals are somewhat conservative (wider than necessary). For safety-stock purposes, over-coverage errs on the side of protection but implies extra holding cost.

Table 5. Probabilistic metrics for the Kalman models. Best value per column in green.

Model	PL Q10 ↓	PL Q50 ↓	PL Q90 ↓	80% coverage
Kalman LL Seasonal	0.1318	**0.5004**	0.3306	85.72%
Kalman LT Seasonal	0.1347	0.5015	**0.3304**	85.43%
Kalman Local Level	**0.1315**	0.5053	0.3366	85.64%
Kalman Single-SKU (M6)	0.1426	0.5058	0.3518	**81.08%**
Kalman Local Trend	0.1419	0.5112	0.3371	85.37%



(a) Empirical coverage of the nominal 80% interval. **(b)** Pinball loss at the 10th, 50th and 90th percentiles.

Figure 4. Probabilistic performance of the Kalman variants (LL = local level, LT = local trend).

6.4 Performance by Volume Band

Table 6 reports the best Kalman model (local level + seasonal) by volume band. The cohort is balanced (8,960 observations per band), yet relative error rises sharply as volume falls: WAPE is 0.7166 for high-volume series but exceeds 1.0 for medium (1.1136) and low volume (1.2740), because the denominator $\sum y_i$ is tiny when most days are zero. MAE moves the opposite way (1.9453 → 0.3566) because absolute demand is lower.

Table 6. Kalman local level + seasonal by volume band.

Volume band	Observations	MAE ↓	RMSE ↓	WAPE ↓
High	8,960	1.9453	3.3110	0.7166
Medium	8,960	0.7002	1.1062	1.1136
Low	8,960	0.3566	0.6813	1.2740

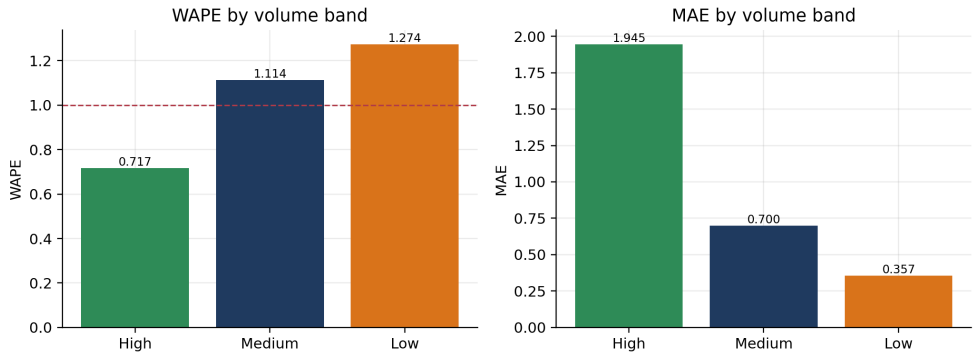


Figure 5. Kalman local level + seasonal: WAPE (left; dashed line marks WAPE = 1) and MAE (right) by volume band.

6.5 Fold-by-Fold Stability

Table 7 and Figure 6 show that the ranking is the same in every fold: LightGBM is best, followed by Kalman LL seasonal, then the naive weekly seasonal and naive last-value baselines. Every model records its highest MAE in the final fold (origin `d1885`), although LightGBM stays below 1.0 throughout (0.8821–0.9679) and the last-value baseline barely changes after fold 1 (1.2314–1.2564). No single fold drives the conclusions.

Table 7. MAE across the four rolling folds. Best per fold in green.

Model	Fold 1801	Fold 1829	Fold 1857	Fold 1885
LightGBM	**0.8821**	**0.9306**	**0.9103**	**0.9679**
Kalman LL Seasonal	0.9723	1.0145	0.9903	1.0257
Naive Weekly Seasonal	1.1460	1.1835	1.1676	1.2507
Naive Last Value	1.2314	1.2528	1.2509	1.2564

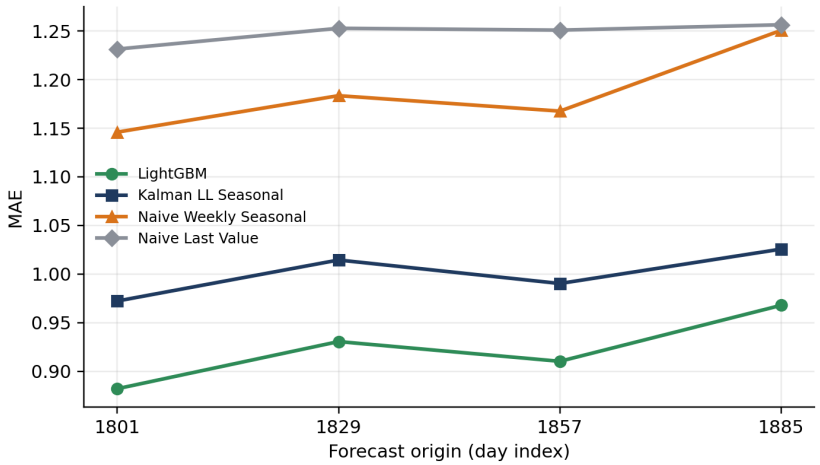


Figure 6. MAE by forecast origin. LightGBM is best at every origin; all models find the last fold hardest.

6.6 Naive Baselines Across Origins

Figure 7 shows the pooled WAPE of the two heuristics on the development origins. The weekly seasonal baseline is at or below the last-value baseline at every origin (roughly 0.97–1.00 vs. 0.99–1.07), confirming that the 7-day cycle carries real signal. Both hover around a WAPE of 1.0, meaning that a naive forecast is barely better than predicting nothing at all — the bar that the learned models must clear.

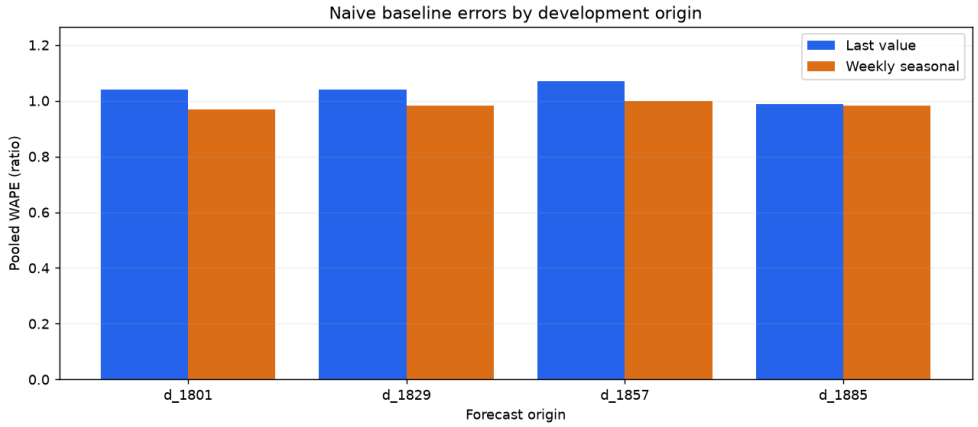
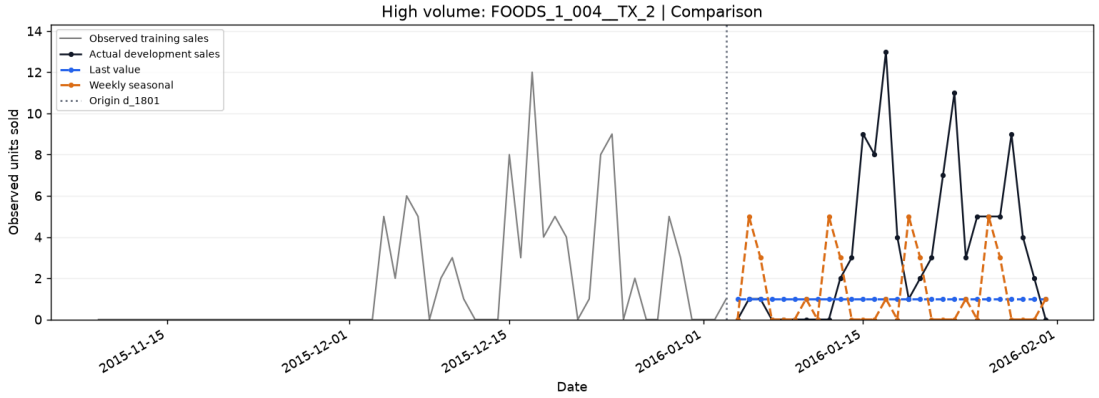


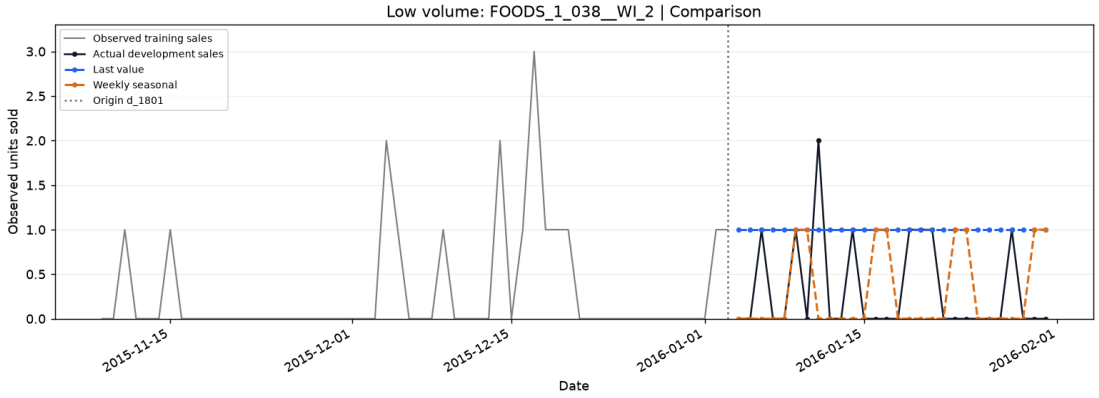
Figure 7. Pooled WAPE of the naive baselines by development origin.

6.7 Qualitative Forecast Examples

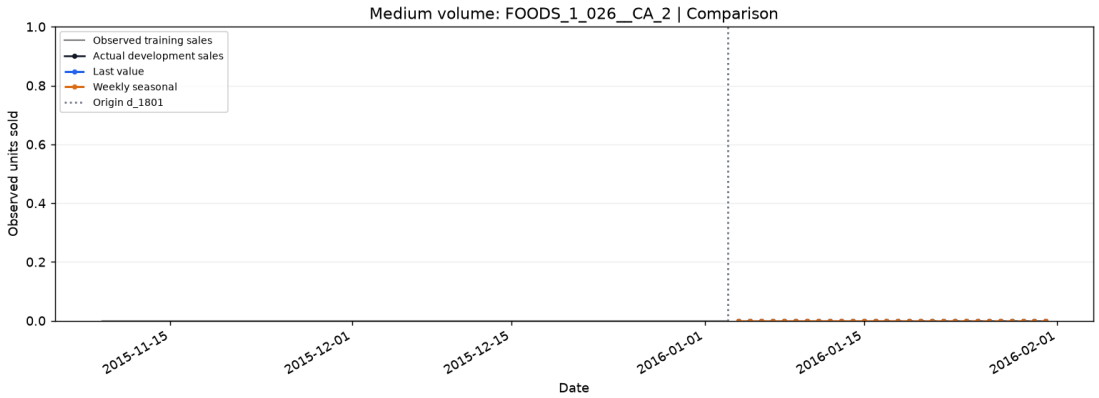
We inspect one representative series per volume band. Figure 8 shows the naive forecasts at origin `d1801`. For the high-volume series `FOODS_1_004__TX_2` the last-value baseline stays flat at 1 unit while actual sales climb in mid-January with spikes up to 13 units, so both baselines miss the peaks by a wide margin; the weekly seasonal baseline reproduces a repeating pattern that does not match the realised timing. For the low-volume series `FOODS_1_038__WI_2` sales are sporadic 0/1 events, and flat or periodic forecasts cannot anticipate which days will be non-zero. The medium-volume example `FOODS_1_026__CA_2` is all zeros in this window, so every baseline is trivially correct there — a reminder of how heavily zeros dominate this cohort.



(a) High volume.



(b) Low volume.



(c) Medium volume.

Figure 8. Naive baselines (last value and weekly seasonal) versus actual sales at origin `d1801`, one series per volume band.

Figure 9 shows the Kalman variants with their q_{10}–q_{90} bands at the last origin (`d1885`). Three behaviours are visible:

- **Median forecasts hug zero.** For intermittent series the q_{50} forecast sits near zero; the seasonal variants add a small weekly ripple (e.g. peaks around 0.4 units for the low-volume series) while the non-seasonal variants are flat.
- **Bands widen with the horizon,** consistent with $V_h = P_{T|T} + hQ + R$, and reach about 1 unit for the low- and medium-volume series.
- **Wide upper bands after long zero runs.** For the high-volume series the realised sales are zero throughout the window, yet the upper band rises to roughly 3–5 units. The model is uncertain because the series was active earlier in the year; here the interval is conservative, which fits the observed over-coverage in Table 5. For the medium-volume series, the series has no sales history in the plotted training window and then records occasional sales of 1–3 units which the median forecast cannot anticipate, although most fall inside the 80% band.

7 Discussion

7.1 Why LightGBM Wins on Point Accuracy

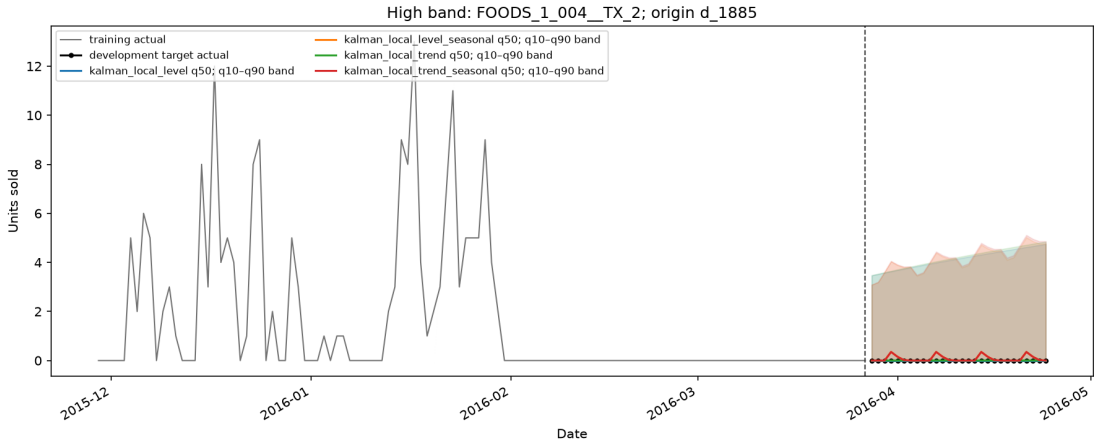
Global pooling lets the model learn calendar, SNAP, price and lag effects from all 240 series at once, whereas each Kalman filter must infer its parameters from a single, mostly-zero series. The L1 objective also targets the median, which suits zero-heavy data and is consistent with the MAE and WAPE metrics on which LightGBM leads. The paper's working hypothesis is that local maximum-likelihood estimation of (Q, R) becomes unreliable when about 89% of days are zero (low-volume band); a per-band breakdown for LightGBM would be needed to confirm this directly.

7.2 Why Kalman Filters Remain Valuable

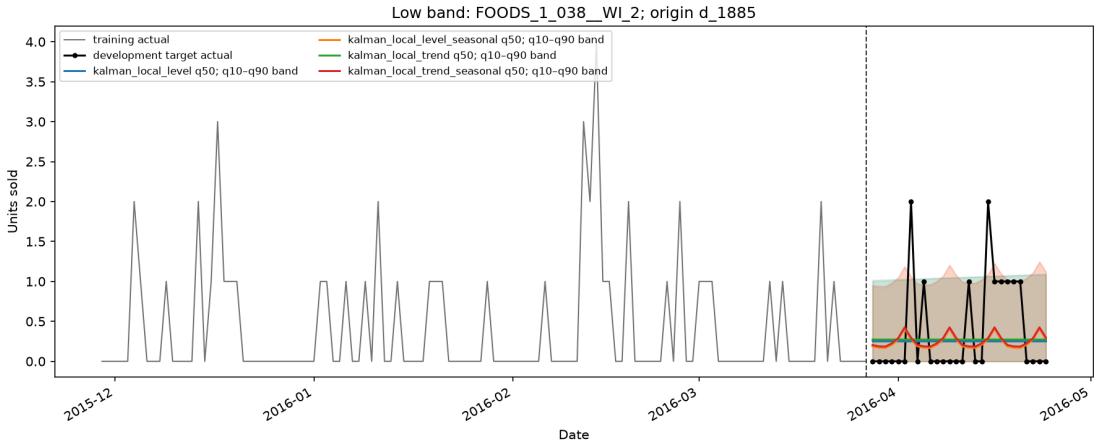
- **Calibrated uncertainty for free:** predictive variance V_h is analytic, and all five variants reach at least the nominal 80% coverage.
- **Interpretability:** the latent level, trend and seasonal states have direct business meaning.
- **Missing-data handling:** the filter skips updates for unobserved days instead of imputing values.
- **Best RMSE:** the seasonal local-level model gives the lowest squared error of all models.

7.3 Limitations

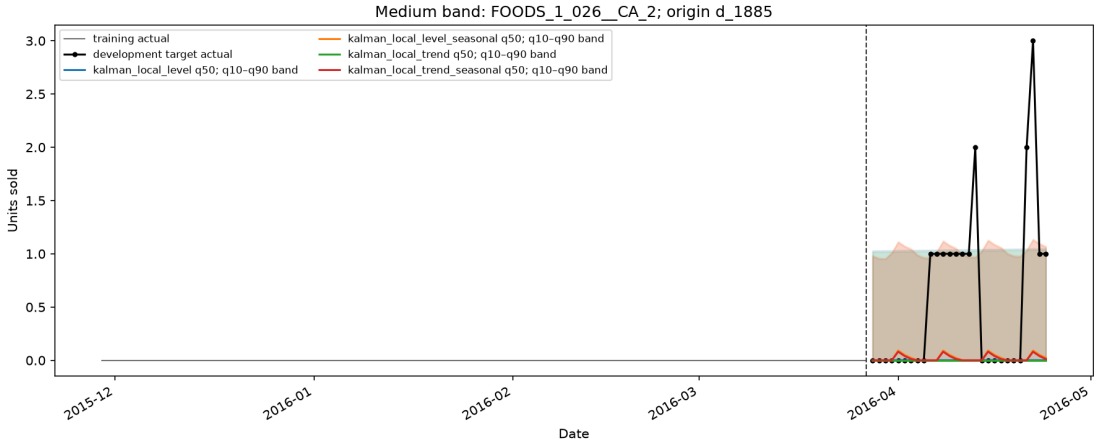
- **Gaussian predictive intervals** are only an approximation for non-negative, count-like, zero-heavy demand; zero-clipping fixes the sign but not the shape of the distribution.
- **Over-coverage** (85–86%) in four of the five variants shows the intervals are not perfectly calibrated and could be tightened.
- **No probabilistic output for LightGBM** in this phase, so the two paradigms cannot yet be compared on pinball loss or coverage. Quantile-objective LightGBM would address this.



(a) High volume (`FOODS_1_004__TX_2`).



(b) Low volume (`FOODS_1_038__WI_2`).



(c) Medium volume (`FOODS_1_026__CA_2`).

Figure 9. Kalman variants at origin `d1885`: median forecast (q_{50}) and 10th–90th percentile band against actual sales.

- **Cohort size:** 240 series and four folds are a small sample of the full M5 hierarchy, and the pooled WAPE for low- and medium-volume series exceeds 1.0 for all models.
- **Exogenous variables:** the Kalman variants here use no SNAP, price or event regressors, which LightGBM does.

8 Interactive Diagnostic Platform

To bridge theory and practice we built a Streamlit application (`app.py`) with three components:

1. **Live Kalman simulation:** sliders for Q and R show the effect on the Kalman gain K_t and on the widening 80% prediction intervals.
2. **Leakage-safe feature explorer:** real-time construction of the 26 tabular features at a chosen forecast origin T.
3. **Interactive leaderboards:** bar charts and metric sorting across all eight models.

9 Conclusion and Operational Recommendations

Neither local state-space models nor a global tree-based forecaster is a complete solution on its own. Our results support a hybrid policy:

1. **Procurement baseline:** use global LightGBM for purchase-order sizing, since it has the lowest expected unit error (WAPE 0.7640, MAE 0.9227).
2. **Safety-stock sizing:** use the Kalman predictive variance V_h to set dynamic, calibrated safety-stock buffers (81.08% empirical coverage for the single-SKU model at a nominal 80%).

Phase 2 roadmap. Future work will combine the two modalities in global probabilistic recurrent architectures such as DeepAR [2] and Temporal Fusion Transformers, using Kalman-filtered states as dynamic structural inputs, and will add quantile-based LightGBM so that all models can be compared on probabilistic metrics.

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